$$\mathcal{L} = \frac{1}{|M|} \sum_{i \in M} \frac{1}{P^2 C} \|\widehat{x}_i - t_i\|_2^2$$

$$t_i = \frac{x_i - \mu_i}{\sqrt{\sigma_i^2 + \epsilon}}, \qquad |M| = 147$$